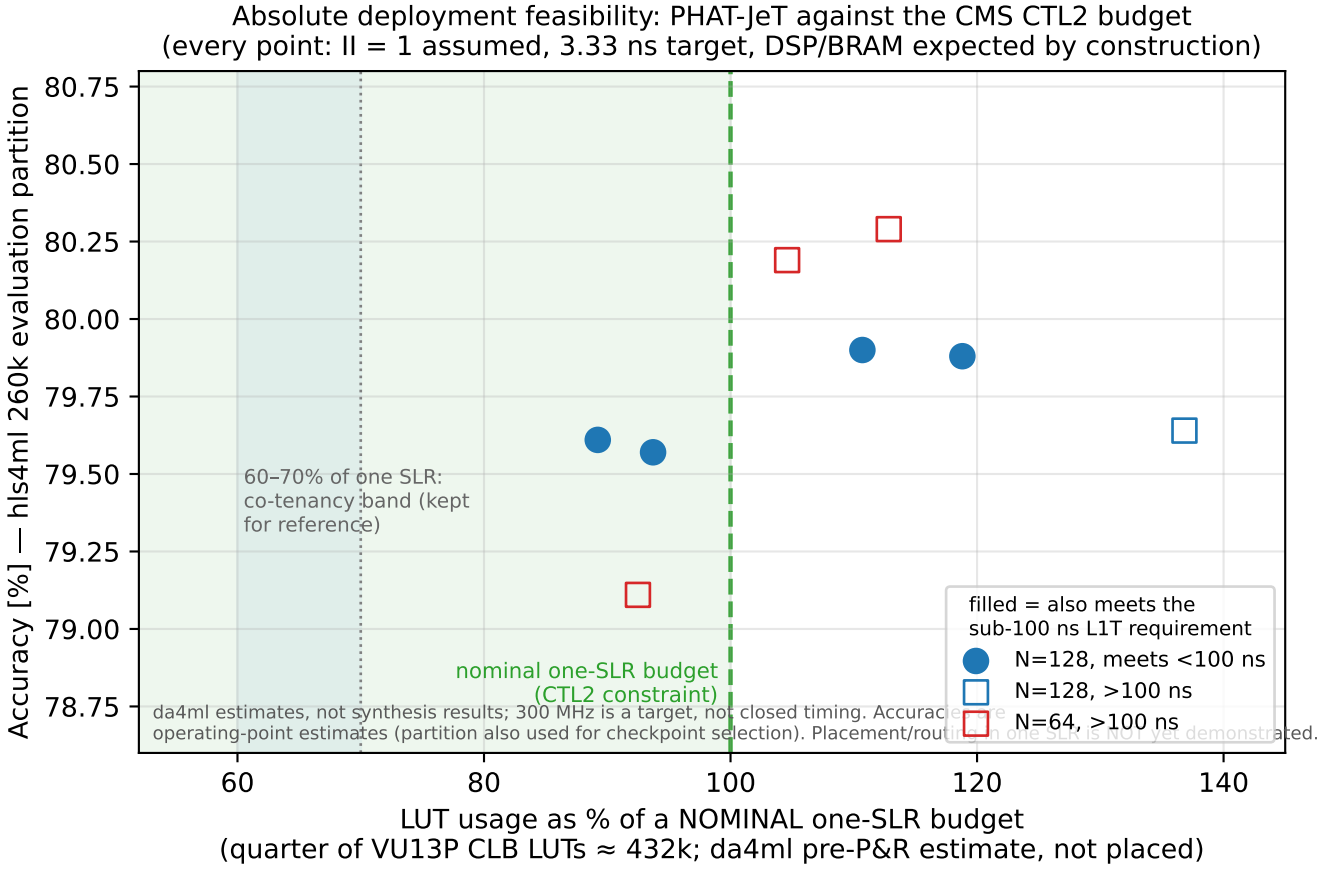
Absolute deployment feasibility: PHAT-JeT against the CMS CTL2 budget
(every point: II = 1 assumed, 3.33 ns target, DSP/BRAM expected by construction)
Accuracy [%] — hls4ml 260k evaluation partition
80.75
80.50
80.25
80.00
79.75
79.50
79.25
79.00
78.75
60–70% of one SLR:
co-tenancy band (kept
for reference)
nominal one-SLR budget
(CTL2 constraint)
filled = also meets the
sub-100 ns L1T requirement
N=128, meets <100 ns
N=128, >100 ns
N=64, >100 ns
da4ml estimates, not synthesis results; 300 MHz is a target, not closed timing. Accuracy are
operating-point estimates (partition also used for checkpoint selection). Placement/routing on one SLR is NOT yet demonstrated.
60
80
100
120
140
LUT usage as % of a NOMINAL one-SLR budget
(quarter of VU13P CLB LUTs ≈ 432k; da4ml pre-P&R estimate, not placed)